

María Isabel Masaya Córdoba

Full Stack Developer

Guatemala City, Guatemala • silverisa17@gmail.com

github.com/isabh17 • isabh17.github.io

Education

Universidad de San Carlos de Guatemala

B.Sc. in Computer Science and Systems Engineering

Guatemala City, Guatemala

Coursework completed: October 2025

Colegio Sagrado Corazón de Jesús

High School Diploma in Sciences and Letters

Guatemala City, Guatemala

2017

Experience

Doctor Online

Full Stack Developer

Guatemala City, Guatemala

2024 – Present

- Develop software for a telemedicine platform operating in over 14 countries with more than 2 million affiliated patients, under HIPAA and GDPR regulations for handling clinical information.
- Build web features on both the patient and physician sides: appointment scheduling, electronic health records, notifications and corporate portals.
- Work across a microservices architecture, contributing to backend services as well as the web applications that consume them.
- Design relational schemas, migrations and stored procedures in MySQL, safeguarding the integrity and traceability of sensitive medical data.
- Optimize SQL queries to keep response times stable as data volume grows, preventing degradation in critical features.
- Provide direct technical support to patients, physicians and corporate clients: translate non-technical reports into concrete causes in the system and coordinate resolution with QA and support through Jira.
- Integrate AI assistants into daily development — code generation, debugging and exploration of existing codebases — always validating and adjusting the output before it reaches production.
- Work with the team toolset: Jira for ticket management and tracking, Bitbucket for version control and code review, Slack for daily coordination, and the internal wiki for technical documentation.
- Take part in the review of technical architecture and product documents alongside the team.
- Produce reports and data analysis with SQL and R / RStudio to support product decisions.

Universidad de San Carlos de Guatemala

Teaching Assistant — Intermediate Practicum

Guatemala City, Guatemala

2023 – 2024

- Taught the lab sessions of the Intermediate Practicum course, explaining course content and resolving students' questions.
- Designed the course assignments and projects, defining specifications and grading criteria.
- Graded assignments, projects and exams, delivering individual feedback to each student.
- Advised students outside class hours, supporting the development of their projects.

Joyería y Relojería El Brillante

Administrative Assistant

Guatemala City, Guatemala

2020 – 2023

- Coordinated the schedule and sales operations of the business, sustaining daily operations while pursuing the engineering degree.
- Served customers directly, in person and by phone, resolving questions and complaints on the spot: the foundation of the communication with non-technical users now applied when providing support as a developer.
- Drafted correspondence, reports and internal documentation, developing precision and clarity in written communication.
- Managed the advertising and social media presence of the business.

Projects

Relational Databases in Oracle

Academic project

Oracle · PL/SQL · Python · Flask

2024

- Designed two database systems from conceptual model through implementation, with 20 tables in total.
- Implemented 10 stored procedures and audit triggers on every table, logging every change made to the data.
- Built bulk loading of 20,000 records from six CSV files through a staging table, respecting foreign-key order.
- Developed a Flask REST API and 25 business queries over the model.

Distributed Voting System

Academic project

Go · gRPC · Kafka · Redis · MongoDB · Kubernetes

2024

- Built a high-volume ingestion pipeline with Go microservices communicating over gRPC.
- Queued traffic with Kafka to decouple ingestion from writes and prevent data loss during load spikes.
- Persisted data in Redis for live counters and MongoDB as the durable record.
- Deployed 6 containerized services on Kubernetes, with Grafana dashboards and Locust load testing.

Neural Network Chatbot

Academic project

TensorFlow · Keras · NLTK · TensorFlow.js

2025

- Trained a Keras neural network to classify 784 intents from 2,075 training patterns.
- Processed natural language with NLTK through tokenization and lemmatization to build the model vocabulary.
- Exported the model to TensorFlow.js to run inference in the browser, removing the need for a server.

Leadership & Activities

Revista ECYS — School of Computer Science and Systems, USAC

Guatemala City, Guatemala

Editorial Design and Layout

2026

- Designed and laid out the 28th edition, 51 pages, of the digital magazine of the School of Computer Science and Systems at the Faculty of Engineering: a biannual publication with twelve years and twenty-nine editions of history.
- Produced the complete edition in R / RStudio, building a reproducible publishing pipeline in which the document is generated from source and layout rules are defined once, rather than applied page by page.
- Defined the typographic hierarchy and grid of the issue, sustaining the readability of technical content on artificial intelligence, data and automation throughout the edition.
- Adapted articles of varying length and density into a consistent visual format, ensuring each piece kept its clarity within the magazine's graphic language.
- Published the edition on Issuu, where it remains permanently and publicly available to the academic community.

Skills & Interests

Technical: Advanced: Python, SQL, MySQL, Oracle, PL/SQL, Git · Intermediate: React, TypeScript, JavaScript, HTML/CSS, Go, Flask, Docker · Project experience: Kubernetes, Kafka, Redis, MongoDB, gRPC, TensorFlow, Keras, R/RStudio · Tools: Jira, Bitbucket, Slack, Confluence

Soft skills: Effective communication with non-technical users · Empathy and service mindset · Teamwork and cross-functional collaboration · Customer service · Active listening · Patience explaining technical topics

Languages: Spanish (native) · English (intermediate)

Interests: Editorial design and typography · Hiking and travel across Guatemala · Baking